

Assignment 2

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Data preparation

I added a few changes to the original file, Java had a memory error, so I had to add

```
.config("spark.driver.memory", "4g")
.config("spark.executor.memory", "2g")
```

and delete

```
.sample(fraction=100 * n / df.count(), seed=0)
```

to spend less memory, after that everything worked and at the end of the code I wrote

```
df.write
.option("sep", "\t")
.mode("overwrite")
.csv("/index/data")
```

to save the id, title and text in /index/data in a file

Indexer

1. mapper1.py

Firstly we split text to words using regex.

Then mapper divides the data into 3 types: docs, terms and const.

- docs contains information related to documents(document id, title and length of the document)
- terms contains information about unique terms(term, document id)
- last category is constants, but finally it has only 1 value(average length of the documents)

To distinguish which data was received at the next stage of the pipeline, the first word is always the type.

Console output for

```
hdfs dfs -cat /index/data/part* — python3 /app/mapreduce/mapper1.py
```

```
hadoop root@cluster-master: /app
terms the 44829245
terms hole 44829245
terms a 44829245
terms child 44829245
terms is 44829245
terms born 44829245
terms 1955 44829245
terms incidental 44829245
terms music 44829245
terms by 44829245
terms bernard 44829245
terms herrmann 44829245
terms see 44829245
terms also 44829245
terms when 44829245
terms a 44829245
terms child 44829245
terms is 44829245
terms born 44829245
terms a 44829245
terms 1974 44829245
terms song 44829245
terms usually 44829245
terms associated 44829245
terms with 44829245
terms christmas 44829245
terms with 44829245
terms a 44829245
terms tune 44829245
terms from 44829245
terms ciro 44829245
terms dammicco 44829245
terms alias 44829245
terms zacar 44829245
terms and 44829245
terms lyrics 44829245
terms from 44829245
terms fred 44829245
terms jay 44829245
terms performed 44829245
terms hv 44829245
```

2. reducer.py

We send the same data to the reducer, but we add the (term frequency) to the terms, which we count in the same file.

Console output for

```
hdfs dfs -cat /index/data/part* — python3 /app/mapreduce/mapper1.py — python3 /app/mapreduce/reducer1.py
```

```
hadoop root@cluster-master: /app
terms jazz 44829245 1
terms standard 44829245 1
terms 1969 44829245 1
terms instrumental 44829245 1
terms thad 44829245 1
terms jones 44829245 1
terms 1993 44829245 1
terms recording 44829245 1
terms choir 44829245 1
terms of 44829245 1
terms trinity 44829245 1
terms college 44829245 1
terms cambridge 44829245 1
terms under 44829245 1
terms richard 44829245 1
terms marlow 44829245 1
terms song 44829245 2
terms brand 44829245 1
terms nubian 44829245 1
terms from 44829245 3
terms the 44829245 2
terms 1997 44829245 1
terms soundtrack 44829245 1
terms soul 44829245 1
terms in 44829245 1
terms hole 44829245 1
terms 1955 44829245 1
terms incidental 44829245 1
terms bernard 44829245 1
terms herrmann 44829245 1
terms see 44829245 1
terms also 44829245 1
terms when 44829245 1
terms 1974 44829245 1
terms usually 44829245 1
terms associated 44829245 1
terms with 44829245 2
terms tune 44829245 1
terms ciro 44829245 1
terms dammicco 44829245 1
terms alias 44829245 1
```

3. index.sh

take initial data from /index/data.

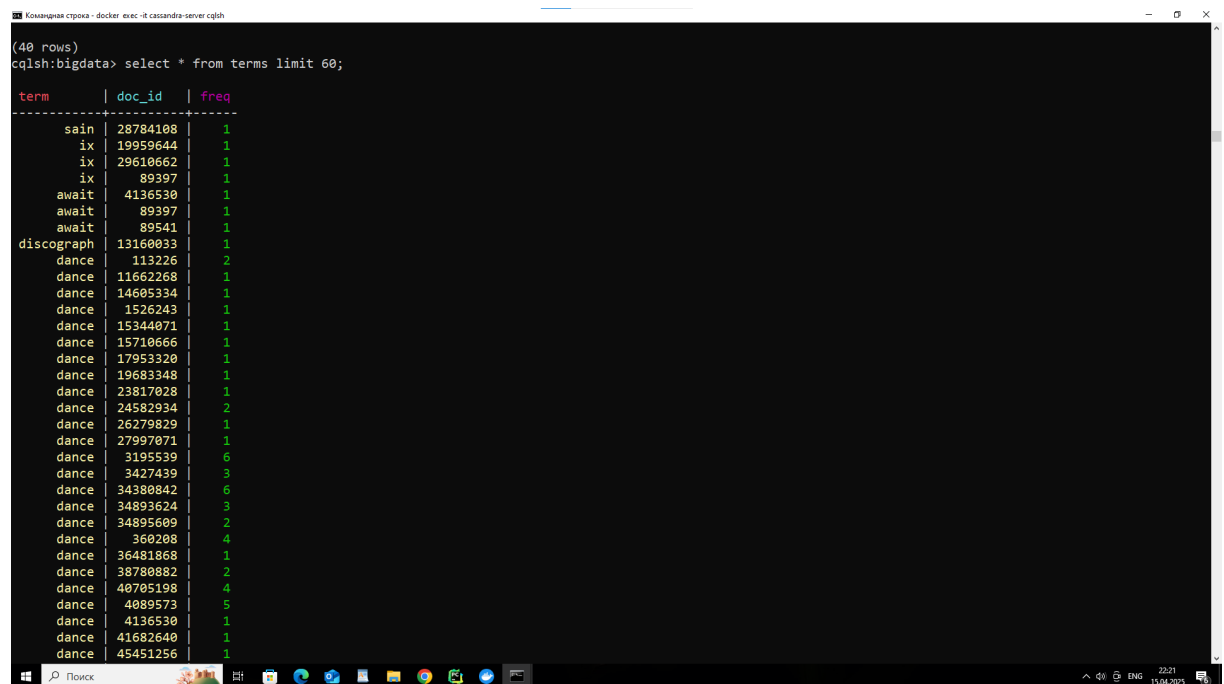
use /tmp/index/output1 as temporary.

4. app.py

create 3 tables(terms, docs and constants).

- docs table has columns (doc_id TEXT, title TEXT, len INT) with Primary key (doc_id)
- terms table has columns (term TEXT, doc_id TEXT, freq INT) with Primary key (term, doc_id)
- constants table has columns (var TEXT, value INT) with Primary key (var)

final tables:



```
(40 rows)
cqlsh:bigdata> select * from terms limit 60;
```

term	doc_id	freq
sain	28784108	1
ix	19959644	1
ix	29610662	1
ix	89397	1
await	4136530	1
await	89397	1
await	89541	1
discograph	13160033	1
dance	113226	2
dance	11662268	1
dance	14605334	1
dance	1526243	1
dance	15344071	1
dance	15710666	1
dance	17953320	1
dance	19683348	1
dance	23817028	1
dance	24582934	2
dance	26279829	1
dance	27997071	1
dance	3195539	6
dance	3427439	3
dance	34380842	6
dance	34803624	3
dance	34805600	2
dance	360208	4
dance	36401868	1
dance	38780882	2
dance	40705198	4
dance	4089573	5
dance	4136530	1
dance	41682640	1
dance	45451256	1

Big Data

```
Konstantin@cpkova:~$ docker exec -it cassandra-server cqlsh
(60 rows)
cqlsh:bigdata> select * from docs limit 60;

doc_id | len | title
-----|----|-----
33456414 | 695 | A Causal Theory of Knowing
38294693 | 338 | A Change Is Gonna Come (Jack McDuff album)
68147671 | 260 | A Beast the Color of Winter
1719074 | 1675 | A Case of Conscience
61110799 | 221 | A Boy, a Girl and a Dog
5048088 | 1334 | A Case of Identity
195945 | 4546 | A Canticle for Leibowitz
8100298 | 1271 | A Battle of Wits (2006 film)
45349228 | 180 | A Big Country
2301932 | 1503 | A Bar at the Folies-Bergère
73781068 | 239 | A Breeze of Hope
6164295 | 462 | A Bewitched Place
69610272 | 450 | A Bilingual Field Guide to the Frogs of Zululand
39571443 | 364 | A Bunch of Munsch
7125736 | 587 | A Body in the Bath House
56986411 | 734 | A Brand New Me (Aretha Franklin album)
21875355 | 81 | A Balloon Called Moaning
21950055 | 438 | A Bunch of Amateurs
25554252 | 243 | A Brief History of Amazing Letdowns
16370814 | 458 | A Breathtaking Guy
3307869 | 921 | A Breath of Snow and Ashes
36350251 | 617 | A Capitalism for the People
51090748 | 260 | A Cartload of Clay
54735192 | 219 | A Better World (album)
50702386 | 278 | A Brilliant Genocide
39684157 | 181 | A Change of Heart (Bernard Butler song)
24667353 | 523 | A Brief History of Anxiety (Yours & Mine)
17053320 | 392 | A Bit of Fry & Laurie
65242707 | 904 | A Better Time
4554788 | 614 | A Boy's Own Story
37729193 | 569 | A Certain Slant of Light (novel)
51124146 | 682 | A C Bertelli
56910921 | 642 | A Bride for Rip Van Winkle
34879342 | 134 | A Celtic Christmas
34893624 | 837 | A Case of You (film)
32729135 | 1283 | A Cartoonist's Nightmare
30449058 | 367 | A Burning Hot Summer
47477008 | 24 | A Beautiful Soul
28182107 | 3182 | A Change Is Gonna Come (Grey's Anatomy)
58622515 | 840 | A Boy and a Priest
30449058 | 367 | A Burning Hot Summer
47477008 | 24 | A Beautiful Soul
28182107 | 3182 | A Change Is Gonna Come (Grey's Anatomy)
58622515 | 840 | A Boy and a Priest
30449058 | 367 | A Burning Hot Summer
47477008 | 24 | A Beautiful Soul
28182107 | 3182 | A Change Is Gonna Come (Grey's Anatomy)
58622515 | 840 | A Boy and a Priest
62485656 | 663 | A Calf for Christmas
64954245 | 85 | A Broke Moon Rises
16564749 | 125 | A Aquarii
43752912 | 1011 | A Ballad upon the Popish Plot
60718335 | 220 | A Automobile Company
49650686 | 356 | A 250 Years Old Person
69185349 | 1290 | A Case of You (horse)
12582138 | 141 | A Benefit for Victims of Violent Crime
63351054 | 144 | A Brief Introduction to Unnatural Lightyears
61956180 | 238 | A Calculated Risk
13411302 | 85 | A Celtic Renaissance Wedding
53577759 | 284 | A Bag of Marbles (1975 film)
4241755 | 1200 | A Bullet for the General
58299136 | 322 | A Borrowed Life
46439777 | 208 | A Broken Heart Can Mend
59959284 | 319 | A Cabinet of Curiosities (painting)
28380942 | 227 | A & P Food Stores Building
36612945 | 160 | A Cassette Tape Culture (Phase Two)
58552463 | 204 | A Campaign and Adventure Guidebook for Middle-earth
9848866 | 268 | A Big 10-8 Place

(60 rows)
cqlsh:bigdata> select * from constants;

var | value
----|-----
avg | 566

(1 rows)
cqlsh:bigdata>
```

Ranker

Using the tables, we calculate the bm25 for each pair (term, document), knowing this, we can calculate the bm25 for each pair (term, query) and take the 10 best results.

Query: "This is a query"

```
root@cluster-master: /app
25/04/15 19:46:53 INFO DAGScheduler: Job 14 is finished. Cancelling potential speculative or zombie tasks for this job
25/04/15 19:46:53 INFO TaskSchedulerImpl: Killing all running tasks in stage 19: Stage finished
25/04/15 19:46:53 INFO DAGScheduler: Job 14 finished: count at NativeMethodAccessorImpl.java:0, took 0.016173 s
25/04/15 19:46:53 INFO V2ScanRelationPushDown:
Pushing operators to terms:
Pushed Filters: EqualTo(term,is)
Post-Scan Filters:

25/04/15 19:46:53 INFO V2ScanRelationPushDown:
Output: term#4, doc_id#5, freq#6

25/04/15 19:46:53 INFO SparkContext: Starting job: collect at /app/query.py:43
25/04/15 19:46:53 INFO DAGScheduler: Got job 15 (collect at /app/query.py:43) with 1 output partitions
25/04/15 19:46:53 INFO DAGScheduler: Final stage: ResultStage 20 (collect at /app/query.py:43)
25/04/15 19:46:53 INFO DAGScheduler: Parents of final stage: List()
25/04/15 19:46:53 INFO DAGScheduler: Missing parents: List()
25/04/15 19:46:53 INFO DAGScheduler: Submitting ResultStage 20 (MapPartitionsRDD[61] at collect at /app/query.py:43), which has no missing parents
25/04/15 19:46:53 INFO MemoryStore: Block broadcast_15 stored as values in memory (estimated size 16.1 KiB, free 366.0 MiB)
25/04/15 19:46:53 INFO MemoryStore: Block broadcast_15_piece0 stored as bytes in memory (estimated size 7.8 KiB, free 365.9 MiB)
25/04/15 19:46:53 INFO BlockManagerInfo: Added broadcast_15_piece0 in memory on cluster-master:37077 (size: 7.8 KiB, free: 366.2 MiB)
25/04/15 19:46:53 INFO SparkContext: Created broadcast 15 from broadcast at DAGScheduler.scala:1585
25/04/15 19:46:53 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 20 (MapPartitionsRDD[61] at collect at /app/query.py:43) (first 15 tasks are for partitions Vector(0))
25/04/15 19:46:53 INFO TaskSchedulerImpl: Adding task set 20.0 with 1 tasks resource profile 0
25/04/15 19:46:53 INFO TaskSetManager: Starting task 0.0 in stage 20.0 (TID 21) (cluster-master, executor driver, partition 0, ANY, 10860 bytes)
25/04/15 19:46:53 INFO Executor: Running task 0.0 in stage 20.0 (TID 21)
25/04/15 19:46:53 INFO Executor: Finished task 0.0 in stage 20.0 (TID 21), 15405 bytes result sent to driver
25/04/15 19:46:53 INFO TaskSetManager: Finished task 0.0 in stage 20.0 (TID 21) in 20 ms on cluster-master (executor driver) (1/1)
25/04/15 19:46:53 INFO TaskSchedulerImpl: Removed TaskSet 20.0, whose tasks have all completed, from pool
25/04/15 19:46:53 INFO DAGScheduler: ResultStage 20 (collect at /app/query.py:43) finished in 0.026 s
25/04/15 19:46:53 INFO DAGScheduler: Job 15 is finished. Cancelling potential speculative or zombie tasks for this job
25/04/15 19:46:53 INFO TaskSchedulerImpl: Killing all running tasks in stage 20: Stage finished
25/04/15 19:46:53 INFO DAGScheduler: Job 15 finished: collect at /app/query.py:43, took 0.030623 s
1) 47515595 A Canine Sherlock Holmes
2) 35717359 A Beak o' Newcassell Sangs
3) 3537854 A Cañiza
4) 3401447 A Ceremony of Carols
5) 69610272 A Bilingual Field Guide to the Frogs of Zululand
6) 34879342 A Celtic Christmas
7) 20574008 A Billion Hands Concert
8) 7068424 A Black Mass
9) 57270816 A Book of American Martyrs
10) 48620807 A Beautiful Life (album)
25/04/15 19:46:53 INFO BlockManagerInfo: Removed broadcast_8_piece0 on cluster-master:37077 in memory (size: 5.9 KiB, free: 366.2 MiB)
25/04/15 19:46:53 INFO SparkContext: Invoking stop() from shutdown hook
25/04/15 19:46:53 INFO SparkContext: SparkContext is stopping with exitCode 0.
25/04/15 19:46:53 INFO BlockManagerInfo: Removed broadcast_6_piece0 on cluster-master:37077 in memory (size: 7.8 KiB, free: 366.2 MiB)
```

Query: "Do you love docker"

```
root@cluster-master: /app
25/04/15 19:50:20 INFO ShuffleBlockFetcherIterator: Started 0 remote fetches in 0 ms
25/04/15 19:50:20 INFO Executor: Finished task 0.0 in stage 19.0 (TID 20), 3995 bytes result sent to driver
25/04/15 19:50:20 INFO TaskSetManager: Finished task 0.0 in stage 19.0 (TID 20) in 8 ms on cluster-master (executor driver) (1/1)
25/04/15 19:50:20 INFO TaskSchedulerImpl: Removed TaskSet 19.0, whose tasks have all completed, from pool
25/04/15 19:50:20 INFO DAGScheduler: ResultStage 19 (count at NativeMethodAccessorImpl.java:0) finished in 0.015 s
25/04/15 19:50:20 INFO DAGScheduler: Job 14 is finished. Cancelling potential speculative or zombie tasks for this job
25/04/15 19:50:20 INFO TaskSchedulerImpl: Killing all running tasks in stage 19: Stage finished
25/04/15 19:50:20 INFO DAGScheduler: Job 14 finished: count at NativeMethodAccessorImpl.java:0, took 0.018906 s
25/04/15 19:50:20 INFO V2ScanRelationPushDown:
Pushing operators to terms:
Pushed Filters: EqualTo(term,love)
Post-Scan Filters:

25/04/15 19:50:20 INFO V2ScanRelationPushDown:
Output: term#4, doc_id#5, freq#6

25/04/15 19:50:20 INFO SparkContext: Starting job: collect at /app/query.py:43
25/04/15 19:50:20 INFO DAGScheduler: Got job 15 (collect at /app/query.py:43) with 1 output partitions
25/04/15 19:50:20 INFO DAGScheduler: Final stage: ResultStage 20 (collect at /app/query.py:43)
25/04/15 19:50:20 INFO DAGScheduler: Parents of final stage: List()
25/04/15 19:50:20 INFO DAGScheduler: Missing parents: List()
25/04/15 19:50:20 INFO DAGScheduler: Submitting ResultStage 20 (MapPartitionsRDD[61] at collect at /app/query.py:43), which has no missing parents
25/04/15 19:50:20 INFO MemoryStore: Block broadcast_15 stored as values in memory (estimated size 16.1 KiB, free 366.2 MiB)
25/04/15 19:50:20 INFO MemoryStore: Block broadcast_15_piece0 stored as bytes in memory (estimated size 7.8 KiB, free 366.2 MiB)
25/04/15 19:50:20 INFO BlockManagerInfo: Added broadcast_15_piece0 in memory on cluster-master:42695 (size: 7.8 KiB, free: 366.3 MiB)
25/04/15 19:50:20 INFO SparkContext: Created broadcast 15 from broadcast at DAGScheduler.scala:1585
25/04/15 19:50:20 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 20 (MapPartitionsRDD[61] at collect at /app/query.py:43) (first 15 tasks are for partitions Vector(0))
25/04/15 19:50:20 INFO TaskSetManager: Starting task 0.0 in stage 20.0 (TID 21) (cluster-master, executor driver, partition 0, ANY, 10860 bytes)
25/04/15 19:50:20 INFO Executor: Running task 0.0 in stage 20.0 (TID 21)
25/04/15 19:50:20 INFO Executor: Finished task 0.0 in stage 20.0 (TID 21), 4570 bytes result sent to driver
25/04/15 19:50:20 INFO TaskSetManager: Finished task 0.0 in stage 20.0 (TID 21) in 18 ms on cluster-master (executor driver) (1/1)
25/04/15 19:50:20 INFO TaskSchedulerImpl: Removed TaskSet 20.0, whose tasks have all completed, from pool
25/04/15 19:50:20 INFO DAGScheduler: ResultStage 20 (collect at /app/query.py:43) finished in 0.024 s
25/04/15 19:50:20 INFO DAGScheduler: Job 15 is finished. Cancelling potential speculative or zombie tasks for this job
25/04/15 19:50:20 INFO TaskSchedulerImpl: Killing all running tasks in stage 20: Stage finished
25/04/15 19:50:20 INFO DAGScheduler: Job 15 finished: collect at /app/query.py:43, took 0.027923 s
1) 51341293 A Cheery Soul
2) 5603747 A Bit of Liverpool
3) 6705829 A Brand
4) 11275833 A Bit of What You Fancy
5) 4061002 A Break from the Norm
6) 3195530 A Charlie Brown Valentine
7) 4745279 A Broad Abroad
8) 23727098 A Brief History of Love
9) 12187025 A Bride of the Plains
10) 3926846 A Change Would Do You Good
25/04/15 19:50:20 INFO SparkContext: Invoking stop() from shutdown hook
25/04/15 19:50:20 INFO SparkContext: SparkContext is stopping with exitCode 0.
25/04/15 19:50:20 INFO CassandraConnector: Disconnected from Cassandra cluster.
25/04/15 19:50:20 INFO SerialShutdownHooks: Successfully executed shutdown hook: Clearing session cache for C* connector
```

Query: "Last one"

Big Data

```
root@cluster-master: /app
25/04/15 19:51:23 INFO ShuffleBlockFetcherIterator: Started 0 remote fetches in 0 ms
25/04/15 19:51:23 INFO Executor: Finished task 0.0 in stage 11.0 (TID 14). 3995 bytes result sent to driver
25/04/15 19:51:23 INFO TaskSetManager: Finished task 0.0 in stage 11.0 (TID 14) in 9 ms on cluster-master (executor driver) (1/1)
25/04/15 19:51:23 INFO TaskSchedulerImpl: Removed TaskSet 11.0, whose tasks have all completed, from pool
25/04/15 19:51:23 INFO DAGScheduler: ResultStage 11 (count at NativeMethodAccessorImpl.java:0) finished in 0.016 s
25/04/15 19:51:23 INFO DAGScheduler: Job 8 is finished. Cancelling potential speculative or zombie tasks for this job
25/04/15 19:51:23 INFO TaskSchedulerImpl: Killing all running tasks in stage 11: Stage finished
25/04/15 19:51:23 INFO DAGScheduler: Job 8 finished: count at NativeMethodAccessorImpl.java:0, took 0.021588 s
25/04/15 19:51:23 INFO V2ScanRelationPushDown:
Pushing operators to terms
Pushed Filters: EqualTo(term,one)
Post-Scan Filters:
25/04/15 19:51:23 INFO V2ScanRelationPushDown:
Output: term#4, doc_id#5, freq#6
25/04/15 19:51:23 INFO SparkContext: Starting job: collect at /app/query.py:43
25/04/15 19:51:23 INFO DAGScheduler: Got job 9 (collect at /app/query.py:43) with 1 output partitions
25/04/15 19:51:23 INFO DAGScheduler: Final stage: ResultStage 12 (collect at /app/query.py:43)
25/04/15 19:51:23 INFO DAGScheduler: Parents of final stage: List()
25/04/15 19:51:23 INFO DAGScheduler: Missing parents: List()
25/04/15 19:51:23 INFO DAGScheduler: Submitting ResultStage 12 (MapPartitionsRDD[39] at collect at /app/query.py:43), which has no missing parents
25/04/15 19:51:23 INFO MemoryStore: Block broadcast_9 stored as values in memory (estimated size 16.1 KiB, free 366.1 MiB)
25/04/15 19:51:23 INFO MemoryStore: Block broadcast_9_piece0 stored as bytes in memory (estimated size 7.8 KiB, free 366.1 MiB)
25/04/15 19:51:23 INFO BlockManagerInfo: Added broadcast_9_piece0 in memory on cluster-master:33947 (size: 7.8 KiB, free: 366.2 MiB)
25/04/15 19:51:24 INFO SparkContext: Created broadcast 9 from broadcast at DAGScheduler.scala:1585
25/04/15 19:51:24 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 12 (MapPartitionsRDD[39] at collect at /app/query.py:43) (first 15 tasks are for partitions Vector(0))
25/04/15 19:51:24 INFO TaskSchedulerImpl: Adding task set 12.0 with 1 tasks resource profile 0
25/04/15 19:51:24 INFO TaskSetManager: Starting task 0.0 in stage 12.0 (TID 15) (cluster-master, executor driver, partition 0, ANY, 10860 bytes)
25/04/15 19:51:24 INFO Executor: Running task 0.0 in stage 12.0 (TID 15)
25/04/15 19:51:24 INFO Executor: Finished task 0.0 in stage 12.0 (TID 15). 7382 bytes result sent to driver
25/04/15 19:51:24 INFO TaskSetManager: Finished task 0.0 in stage 12.0 (TID 15) in 22 ms on cluster-master (executor driver) (1/1)
25/04/15 19:51:24 INFO TaskSchedulerImpl: Removed TaskSet 12.0, whose tasks have all completed, from pool
25/04/15 19:51:24 INFO DAGScheduler: ResultStage 12 (collect at /app/query.py:43) finished in 0.030 s
25/04/15 19:51:24 INFO DAGScheduler: Job 9 is finished. Cancelling potential speculative or zombie tasks for this job
25/04/15 19:51:24 INFO TaskSchedulerImpl: Killing all running tasks in stage 12: Stage finished
25/04/15 19:51:24 INFO DAGScheduler: Job 9 finished: collect at /app/query.py:43, took 0.034514 s
1) 37253684 A Birthday Hansel
2) 5352247 A Chaos of Desire
3) 5481481 A Boy in France
4) 16378814 A Breathtaking Guy
5) 28380942 A & P Food Stores Building
6) 12187025 A Bride of the Plains
7) 11662268 A Bear for Punishment
8) 10106530 A Blank in the Weather Map
9) 62695113 A Beautifully Foolish Endeavor
10) 13074637 A Brilliant Mistake
25/04/15 19:51:24 INFO SparkContext: Invoking stop() from shutdown hook
25/04/15 19:51:24 INFO SparkContext: SparkContext is stopping with exitCode 0.
25/04/15 19:51:24 INFO CassandraConnector: Disconnected from Cassandra cluster.
25/04/15 19:51:24 INFO SerialShutdownHooks: Successfully executed shutdown hook: Clearing session cache for C* connector
```